

Deadlock Conditions (Coffman)

Deadlock requires ALL 4 conditions simultaneously

- Mutual Exclusion – resource held exclusively (lock)
- Hold and Wait – holding one lock while waiting for another
- No Preemption – lock cannot be taken away; must be released voluntarily
- Circular Wait – Thread A waits for B; B waits for A

Prevention – break any one condition

- Lock Ordering – always acquire locks in same global order (break circular w
- tryLock with timeout – break Hold and Wait (release held locks on timeout)
- Use single lock for multiple resources where possible (break Hold and Wait)

```
// Prevention: consistent lock ordering (always low ID first)
public void transfer(Account from, Account to, BigDecimal amount) {
    Account first = from.getId() < to.getId() ? from : to;
    Account second = from.getId() < to.getId() ? to : from;
    synchronized (first) {
        synchronized (second) {
            from.debit(amount); to.credit(amount);
        }
    }
}

// Detection: JVM thread dump
jstack <pid> # look for "Found one Java-level deadlock" in output
```